## A10.3 Default signal values

This specification suggests that, in general, for maximum IP reuse, an AXI component interface includes all signals. The presence of all signals reduces the risk of error at the system integration phase of the design flow and it can also help support some design flows that do not effectively support default values for absent signals.

The following tables show the AXI required and optional signals, and the default signals values that apply when an optional signal is not implemented:

- Table A10-1 on page A10-113 shows the master interface write channel signals
- Table A10-2 on page A10-114 shows the memory slave interface write channel signals
- Table A10-3 on page A10-115 shows the master interface read channel signals
- Table A10-4 on page A10-116 shows the memory slave interface read channel.

The following sections give more information about the default signal requirements:

- *Master addresses* on page A10-116
- *Slave addresses* on page A10-117
- *Memory slaves* on page A10-117
- *Write transactions* on page A10-117
- *Read transactions* on page A10-117
- *Response signaling* on page A10-117
- *Non-secure and secure accesses* on page A10-117.


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**Table A10-1 Master interface write channel signals and default signal values**

| Signal | Description | Direction | Required? | Default |
|---|---|---|---|---|
| **ACLK** | Global clock | Input | Required | - |
| **ARESETn** | Global reset | Input | Required | - |
| **AWID** | Write address ID | Output | Optional | All zeros |
| **AWADDR** | Write address | Output | Required | - |
| **AWREGION** | Write region | Output | Optional | All zeros |
| **AWLEN** | Burst length | Output | Optional | All zeros, Length 1 |
| **AWSIZE** | Burst size | Output | Optional | Data bus width |
| **AWBURST** | Burst type | Output | Optional | `0b01`, INCR |
| **AWLOCK** | Lock type | Output | Optional | All zeros, Normal access |
| **AWCACHE** | Cache type | Output | Optional | `0b0000` |
| **AWPROT** | Protection type | Output | Required | - |
| **AWQOS** | QoS value | Output | Optional | `0b0000` |
| **AWVALID** | Write address valid | Output | Required | - |
| **AWREADY** | Write address ready | Input | Required | - |
| **WDATA** | Write data | Output | Required | - |
| **WSTRB** | Write strobes | Output | Optional | All ones |
| **WLAST** | Write last | Output | Required | - |
| **WVALID** | Write valid | Output | Required | - |
| **WREADY** | Write ready | Input | Required | - |
| **BID** | Response ID | Input | Optional | - |
| **BRESP** | Write response | Input | Optional | - |
| **BVALID** | Write response valid | Input | Required | - |
| **BREADY** | Response ready | Output | Required | - |


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**Table A10-2 Memory slave interface write channel signals and default signal values**

| Signal name | Description | Direction | Required? | Default |
|---|---|---|---|---|
| **ACLK** | Global clock | Input | Required | - |
| **ARESETn** | Global reset | Input | Required | - |
| **AWID** | Write address ID | Input | Required | - |
| **AWADDR** | Write address | Input | Required | - |
| **AWREGION** | Write region | Input | Optional | - |
| **AWLEN** | Burst length | Input | Required | - |
| **AWSIZE** | Burst size | Input | Required | - |
| **AWBURST** | Burst type | Input | Required | - |
| **AWLOCK** | Lock type | Input | Optional | - |
| **AWCACHE** | Cache type | Input | Optional | - |
| **AWPROT** | Protection type | Input | Optional | - |
| **AWQOS** | QoS value | Input | Optional | - |
| **AWVALID** | Write address valid | Input | Required | - |
| **AWREADY** | Write address ready | Output | Required | - |
| **WDATA** | Write data | Input | Required | - |
| **WSTRB** | Write strobes | Input | Required | - |
| **WLAST** | Write last | Input | Optional | - |
| **WVALID** | Write valid | Input | Required | - |
| **WREADY** | Write ready | Output | Required | - |
| **BID** | Response ID | Output | Required | - |
| **BRESP** | Write response | Output | Optional | `0b00`, OKAY |
| **BVALID** | Write response valid | Output | Required | - |
| **BREADY** | Response ready | Input | Required | - |


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**Table A10-3 Master interface read channel signals and default signals values**

| Signal name | Description | Direction | Required? | Default |
|---|---|---|---|---|
| **ARID** | Read address ID | Output | Optional | All zeros |
| **ARADDR** | Read address | Output | Required | - |
| **ARREGION** | Read region | Output | Optional | `0x0` |
| **ARLEN** | Burst length | Output | Optional | All zeros, Length 1 |
| **ARSIZE** | Burst size | Output | Optional | Data bus width |
| **ARBURST** | Burst type | Output | Optional | `0b01`, INCR |
| **ARLOCK** | Lock type | Output | Optional | All zeros, Normal access |
| **ARCACHE** | Cache type | Output | Optional | `0b0000` |
| **ARPROT** | Protection type | Output | Required | - |
| **ARQOS** | QoS value | Output | Optional | `0b0000` |
| **ARVALID** | Read address valid | Output | Required | - |
| **ARREADY** | Read address ready | Input | Required | - |
| **RID** | Read data ID | Input | Optional | - |
| **RDATA** | Read data | Input | Required | - |
| **RRESP** | Read response | Input | Optional | - |
| **RLAST** | Read last | Input | Optional | - |
| **RVALID** | Read valid | Input | Required | - |
| **RREADY** | Read ready | Output | Required | - |


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**Table A10-4 Memory slave interface read channel signals and default signals values**

| Signal name | Description | Direction | Required? | Default |
|---|---|---|---|---|
| **ARID** | Read address ID | Input | Required | - |
| **ARADDR** | Read address | Input | Required | - |
| **ARREGION** | Read region | Input | Optional | - |
| **ARLEN** | Burst length | Input | Required | - |
| **ARSIZE** | Burst size | Input | Required | - |
| **ARBURST** | Burst type | Input | Required | - |
| **ARLOCK** | Lock type | Input | Optional | - |
| **ARCACHE** | Cache type | Input | Optional | - |
| **ARPROT** | Protection type | Input | Optional | - |
| **ARQOS** | QoS value | Input | Optional | - |
| **ARVALID** | Read address valid | Input | Required | - |
| **ARREADY** | Read address ready | Output | Required | - |
| **RID** | Read data ID | Output | Required | - |
| **RDATA** | Read data | Output | Required | - |
| **RRESP** | Read response | Output | Optional | `0b00`, OKAY |
| **RLAST** | Read last | Output | Required | - |
| **RVALID** | Read valid | Output | Required | - |
| **RREADY** | Read ready | Input | Required | - |

### A10.3.1 Master addresses

**AxADDR** There is no minimum requirement for the number of address bits supplied by a master.

If the system to which the master is connected has a different address bus width than that provided by the master:

- if the system address is wider than is provided by the master then the default value of all zeros must be used for the additional high-order address bits
- if the system address is narrower than is provided by the master then the high-order address bits from the master must be left unconnected.

**Note**

Typically a master supplies 32-bits of addressing, optionally a master can support up to 64-bits of addressing.


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### A10.3.2 Slave addresses

**AxADDR** There is no minimum requirement for the number of address bits used by a slave.

A slave is not required to have low-order address bits to support decoding within the width of the system data bus and can assume that such low-order address bits have a default value of all zeros. If the slave has more address bits than supplied by the interconnect, the higher order address bits use a default value of all zeros.

Typically a memory slave has at least enough address bits to fully decode a 4KB address range.

### A10.3.3 Memory slaves

**AxLOCK** A memory slave is not required to use the **AxLOCK** inputs. However, a memory slave that supports exclusive accesses requires these signals.

**AxCACHE** A memory slave is not required to make use of the **AxCACHE** inputs. A memory slave does not require these signals if either:

- it has no caching behavior
- it caches all transactions in the same way.

### A10.3.4 Write transactions

**WSTRB[3:0]** A master is not required to use the write strobe signals **WSTRB[3:0]** if it always performs full data bus width write transactions. The default value for write strobes is all signals asserted.

**WLAST** A slave is not required to use the **WLAST** signal. Since the length of a write burst is defined, a slave can calculate the last write data transfer from the burst length **AWLEN[7:0]** signals.

### A10.3.5 Read transactions

**RLAST** A master is not required to use the **RLAST** signal. Since the length of a read burst is defined, a master can calculate the last read data transfer from the burst length **ARLEN[7:0]** signals.

### A10.3.6 Response signaling

**RRESP, BRESP** A master does not require the **RRESP** and **BRESP** inputs if it both:

- does not perform exclusive accesses
- does not require notification of transaction errors.

A slave does not require the **RRESP** and **BRESP** outputs if it both:

- does not support exclusive accesses
- does not generate error responses.

### A10.3.7 Non-secure and secure accesses

**AxPROT** A slave that is not required to differentiate between Non-secure and Secure accesses, and that does not require any additional protection support, does not require the **AxPROT** input signals.

**Caution**

Take great care with the **AxPROT** signals. The **AxPROT[1]** signals indicate the Secure or Non-secure nature of the transactions, and incorrect assignment of these bits can lead to incorrect system behavior.


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# Part B

## AMBA AXI4-Lite Interface Specification

# Chapter B1
# **AMBA AXI4-Lite**

This chapter defines the AXI4-Lite interface and associated protocol. AXI4-Lite is suitable for simpler control register-style interfaces that do not require the full functionality of AXI4.

This chapter contains the following sections:


- *Definition of AXI4-Lite* on page B1-122
- *Interoperability* on page B1-124
- *Defined conversion mechanism* on page B1-125
- *Conversion, protection, and detection* on page B1-127.



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